Executive Summary: Exploratory Data Analysis of E-Commerce Orders

DecodeLabs Data Analytics Internship | Project 2: Exploratory Data Analysis | Batch 2026

Headline: A few large orders inflate the average, monthly sales swing sharply, and about 4 in 10 orders end up cancelled or returned.

1. Problem Statement

What do 1,200 e-commerce orders (January 2023 to June 2025) reveal about order value, sales trends, outliers and the relationships between variables, before any dashboard or predictive model is built?

2. Methodology

- **Data:** the cleaned dataset from Project 1 (1,200 rows, 14 columns, no missing values).
- **Descriptive statistics:** count, mean, median and quartiles for Quantity, UnitPrice, ItemsInCart and TotalPrice.
- **Distributions:** histograms of the four numeric columns.
- **Outliers:** boxplot and the IQR method (fences at Q1 - 1.5 x IQR and Q3 + 1.5 x IQR).
- **Categories:** bar charts for OrderStatus, Product, PaymentMethod and ReferralSource.
- **Trend:** total sales per month, shown as a line chart.
- **Relationships:** Pearson correlation heatmap of the numeric columns.
- **Tools:** Python, pandas, matplotlib, seaborn.

3. Key Findings

Finding 1: The typical order is worth about 824, not 1,054

The mean TotalPrice is 1,053.97 but the median is only 823.62, so the data is right-skewed: a few large orders pull the average up, and 59% of orders sit below the mean. Half of all orders fall between 410.52 and 1,578.48, and values range from 11.39 to 3,456.40. Quantity (1 to 5) and UnitPrice (11.39 to 699.93) are each spread fairly evenly, and ItemsInCart is bell-shaped around 5 to 6, yet TotalPrice, which combines them, is skewed.

For reporting a typical order, the median is the more reliable figure.

Finding 2: Only 8 orders (0.67%) are outliers, and they look like genuine big orders

The IQR upper fence is 3,330.41, and 8 orders sit above it, with TotalPrice between 3,334.00 and 3,456.40. All 8 have the maximum Quantity of 5 and a UnitPrice between 666.80 and 691.28, the top of the price range (the maximum is 699.93). TotalPrice equals Quantity x UnitPrice in these rows, so they are real purchases rather than data-entry errors, and they were kept in the data. Together they account for 2.1% of total sales.

Finding 3: 41% of orders are cancelled or returned

Cancelled (250 orders, 20.8%) and Returned (247 orders, 20.6%) are the two largest groups, followed by Pending (237, 19.8%), Shipped (235, 19.6%) and Delivered (231, 19.3%). Cancelled and Returned together are 497 of 1,200 orders (41.4%), while only 231 orders (19.3%), about 1 in 5, are Delivered. This is the biggest business concern in the data.

Finding 4: Monthly sales are volatile, and 2023 was the strongest year

Total sales across the period are 1,264,761.96, an average of 42,158.73 per month (median 39,956.32). The best month was June 2024 (68,068.54) and the weakest was April 2023 (27,751.71). Average monthly sales fell from 46,054 in 2023 to 40,020 in 2024 and 38,647 in January to June 2025, with a rebound to 53,047 in June 2025. With about 40 orders per month (between 27 and 53), a handful of large orders can move a whole month, so individual spikes should not be read as lasting trends.

Finding 5: Orders are spread evenly across products, payment methods and referral sources

Product orders range from 156 (Phone) to 181 (Printer): Printer 181, Tablet 179, Chair 178, Laptop 173, Desk

170, Monitor 163, Phone 156. Online is the most used payment method (258 orders, 21.5%) and Gift Card the least (230, 19.2%), with Cash 246, Credit Card 234 and Debit Card 232 in between. Instagram brings the most orders (259, 21.6%) and Referral the fewest (222, 18.5%), with Email 250, Google 241 and Facebook 228 in between. No single category dominates.

Finding 6: UnitPrice and Quantity drive order value

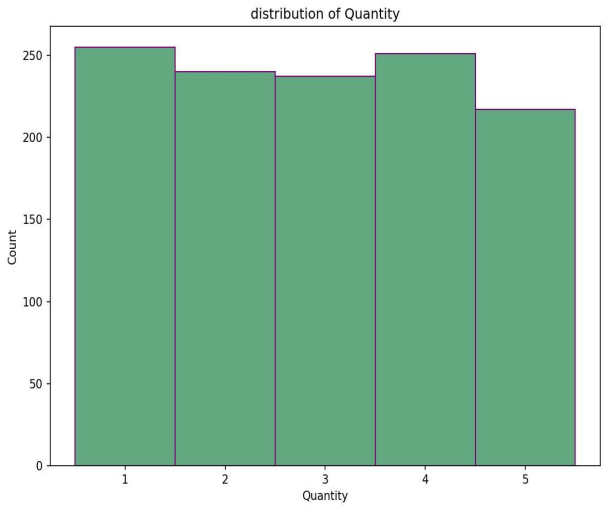
- UnitPrice and TotalPrice: correlation 0.72
- Quantity and TotalPrice: 0.62
- Quantity and ItemsInCart: 0.65
- ItemsInCart and TotalPrice: 0.39
- Quantity and UnitPrice: 0.015, and UnitPrice and ItemsInCart: 0.0006, so these pairs are effectively unrelated

Correlation shows association, not causation.

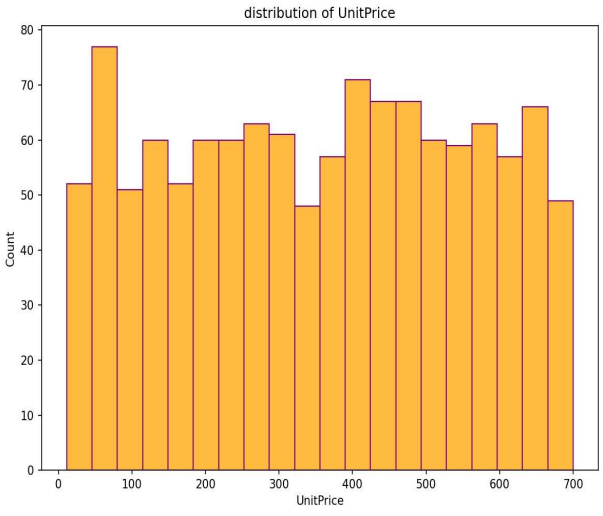
4. Recommendations

1. **Investigate cancellations and returns first.** Break them down by product, payment method and referral source to find where they concentrate.
2. **Report typical order value with the median**, and show the mean alongside it so the effect of large orders stays visible.
3. **Track high-value orders separately** and review their fulfilment, since each one carries a large share of monthly sales.
4. **Find the causes of monthly swings** (promotions, seasonality, campaigns) with more order-level detail, and check what drove the June 2025 rebound.
5. **Look beyond product and channel mix for growth.** Because orders are balanced across categories, the bigger levers appear to be fulfilment and conversion rather than dropping a product or channel.

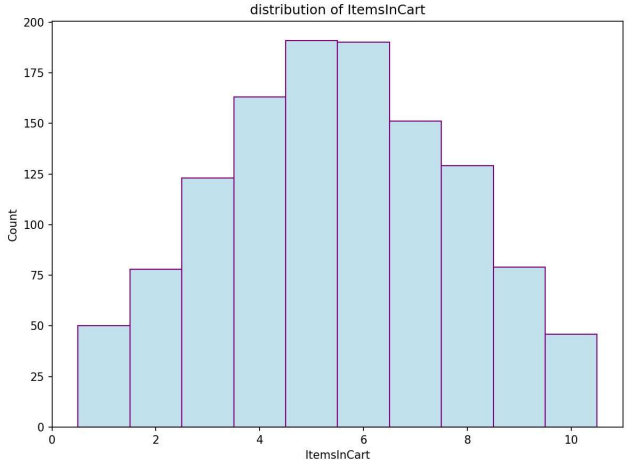
Charts



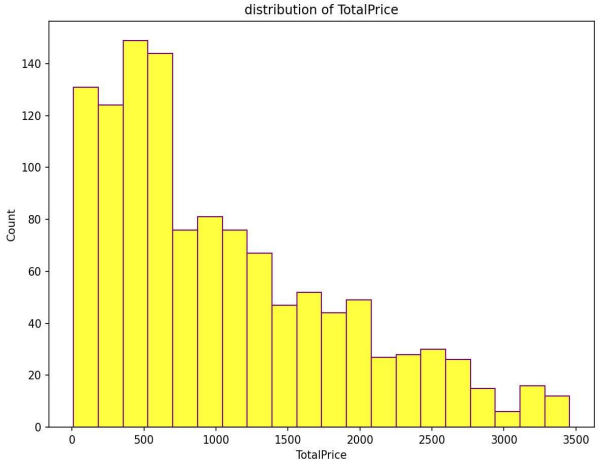
Finding 1: distribution of Quantity



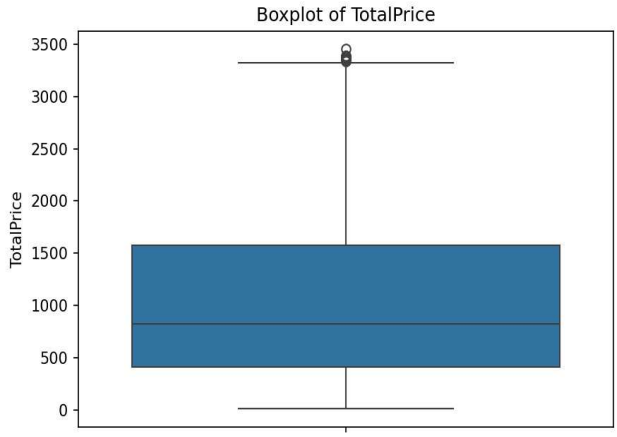
Finding 1: distribution of UnitPrice



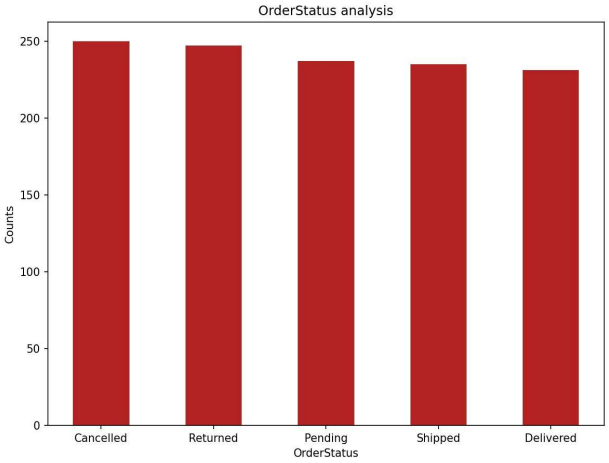
Finding 1: distribution of ItemsInCart



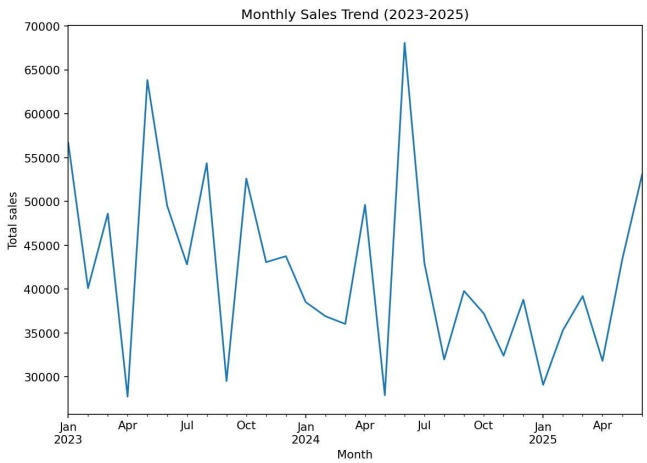
Finding 1: distribution of TotalPrice



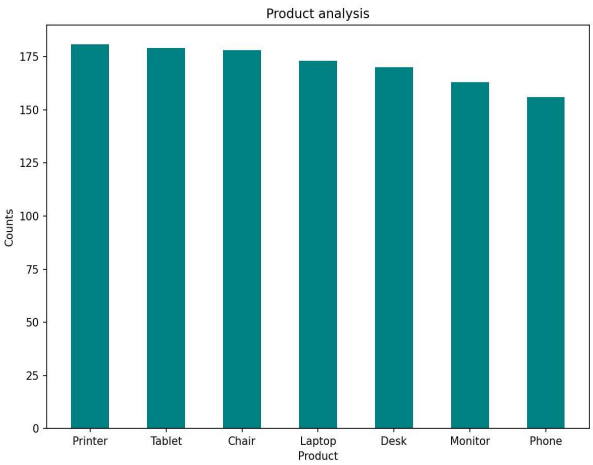
Finding 2: boxplot of TotalPrice



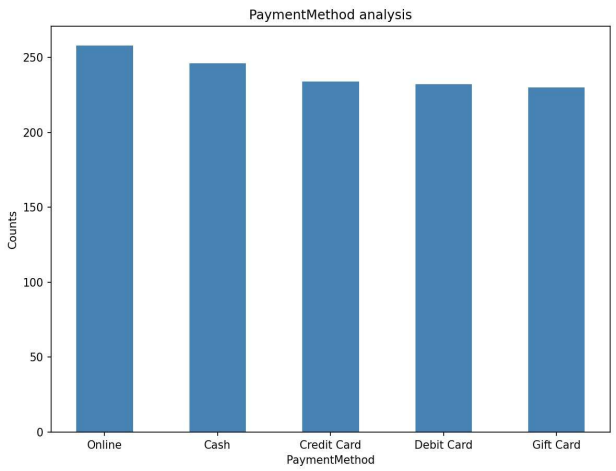
Finding 3: order status



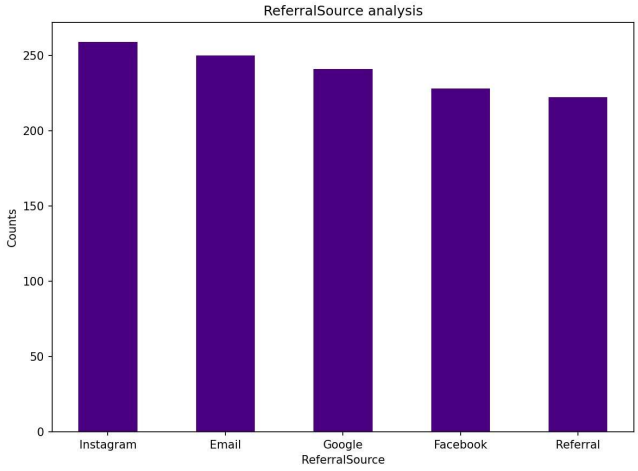
Finding 4: monthly sales trend



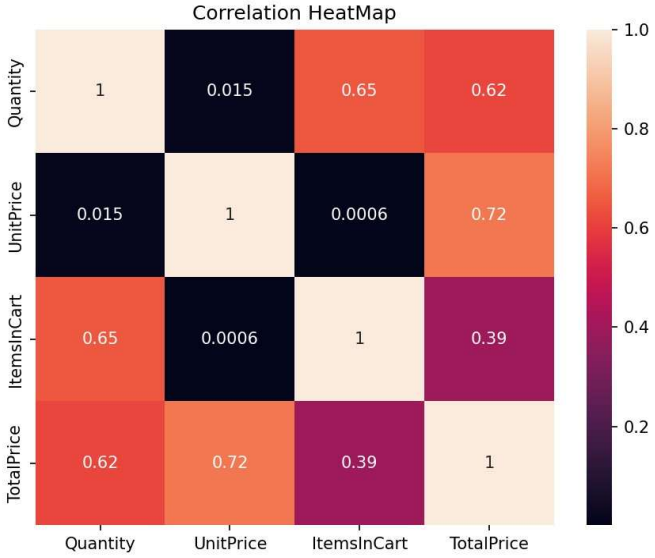
Finding 5: orders by product



Finding 5: orders by payment method



Finding 5: orders by referral source



Finding 6: correlation heatmap